Université d'Ottawa · University of Ottawa

Faculté des sciences
Mathématiques et de statistique

Faculty of Science
Mathematics and Statistics

**MAT 2125, Winter 2023 – Midterm Exam** **Instructor: A. Tikuisis**

Last name: _____

First name: _____

Student number: _____

Read the following instructions:

- The use of cellphones, electronic devices (including calculators), and course notes is strictly forbidden. All phones and electronic devices must be turned off and kept in your bags: do not leave them on you. If you are seen to have an electronic device on your person, we may ask you to leave the exam immediately, and fraud allegations could be made, which could lead to a mark of 0 (zero) on this midterm.
- The duration of this midterm is 75 minutes.
- This is a closed book midterm containing **5 questions** plus one bonus question.
- Do not detach the pages of this test.
- There is an additional blank page at the end of this exam that you may use as scrap paper. If you run out of space, you may use this page or the back of pages. Clearly indicate where to find your answer if it is not entirely contained in the space provided on the page.
- You must give clear and complete solutions, with calculations, explanations and justifications. Make sure that your answer is clearly indicated; you must convince me that you understand your solution in order to receive full marks.

**By signing below, you acknowledge that you are required to respect the above statements.**

*Signature*: _____

THIS SPACE IS RESERVED FOR THE MARKER:

| Question | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|
| Mark | | | | | | |
| Out of | 8 | 8 | 11 | 10 | 10 | 50 |

**1.** Label the following statements as true (T) or false (F). No justification is needed. (1 mark for each correct answer, 0 marks for an incorrect answer or no answer.) (8)

- [ ] If $A \subseteq \mathbb{R}$ is a bounded subset, then $A$ contains a least upper bound.
- [ ] If $(a_n)_{n=1}^{\infty}$ is a sequence that converges, then $(a_n)_{n=1}^{\infty}$ is monotone.
- [ ] If $A \subseteq \mathbb{R}^d$ is an open set and $B \subseteq \mathbb{R}^d$ is a closed set, then $A \setminus B$ is an open set.
- [ ] If $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence in $\mathbb{R}^d$ and we write $a_n = (a_n^{(1)}, \ldots, a_n^{(d)})$ for all $n$, then for $i = 1, \ldots, d$, the sequence $(a_n^{(i)})_{n=1}^{\infty}$ in $\mathbb{R}$ is Cauchy.
- [ ] There exists a set $A \subseteq \mathbb{R}$ such that $\overline{A} = \mathbb{R}$ and $\overline{(\mathbb{R} \setminus A)} = \mathbb{R}$.
- [ ] If $(a_n)_{n=1}^{\infty}$ is a sequence such that $a_n \geq 0$ for all $n$ and $a_n \to 0$ then by the Alternating Series Test, $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ converges.
- [ ] If $(a_n)_{n=1}^{\infty}$ has a subsequence that diverges then $(a_n)_{n=1}^{\infty}$ diverges.
- [ ] There exists a sequence $(a_n)_{n=1}^{\infty}$ such that $a_n \to 0$ and $\sum_{n=1}^{\infty} a_n$ diverges.

**2.** Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers.

(a) Define what it means for $(a_n)_{n=1}^{\infty}$ to **converge** to a number $L \in \mathbb{R}$. (3)

(b) Prove the **Squeeze Theorem**: if $(b_n)_{n=1}^{\infty}, (c_n)_{n=1}^{\infty}$ are other sequences, satisfying

$$a_n \leq b_n \leq c_n \quad \text{for all } n \in \mathbb{N}_{\geq 1},$$

and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$. (5)

**3.** (a) State the **Absolute Convergence Test** for a series, and state what it means for a series to converge **absolutely**. (3)

(b) Give an example of a series that converges, but that does not converge absolutely. Justify why your example has these properties. (4)

(c) Suppose that $(a_n)_{n=1}^{\infty}$ is a bounded sequence. Prove that the series

$$\sum_{n=1}^{\infty} \frac{na_n + 1}{n^3}$$

converges. (4)

**4.** Let $(a_n)_{n=1}^{\infty}$ be a bounded sequence of real numbers.

(a) Give a definition (or equivalent formulation) for

$$\limsup_{n\to\infty} a_n. \tag{3}$$

(b) Compute $\limsup_{n\to\infty} a_n$ where

$$a_n := \begin{cases} \frac{k+1}{k}, & \text{if } n = k! \text{ for some } k \in \mathbb{N}_{\geq 1}; \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

(c) Prove that if $(a_n)_{n=1}^{\infty}$ is a bounded sequence then there exists a subsequence $(a_{n_k})_{k=1}^{\infty}$ that converges to $\limsup_{n\to\infty} a_n$. (4)

**5.** (a) State the definition of what it means for a set $K \subseteq \mathbb{R}^d$ to be (sequentially) compact. (3)

(b) Prove that $K := \{a \in \mathbb{R}^d : \|a\|_2 \leq 1\}$ is compact, from the definition. **Do not use the Heine–Borel theorem.** (5)

**6.** (Bonus) Let $(a_n)_{n=1}^{\infty}$ be a sequence that converges to $L$. For each $n \in \mathbb{N}_{\geq 1}$, define

$$b_n := \frac{a_1 + \cdots + a_n}{n}.$$

Prove that $(b_n)_{n=1}^{\infty}$ also converges to $L$. `(up to 5 bonus marks)`

(the last page)